DMPK Case Studies

PK Bioanalysis

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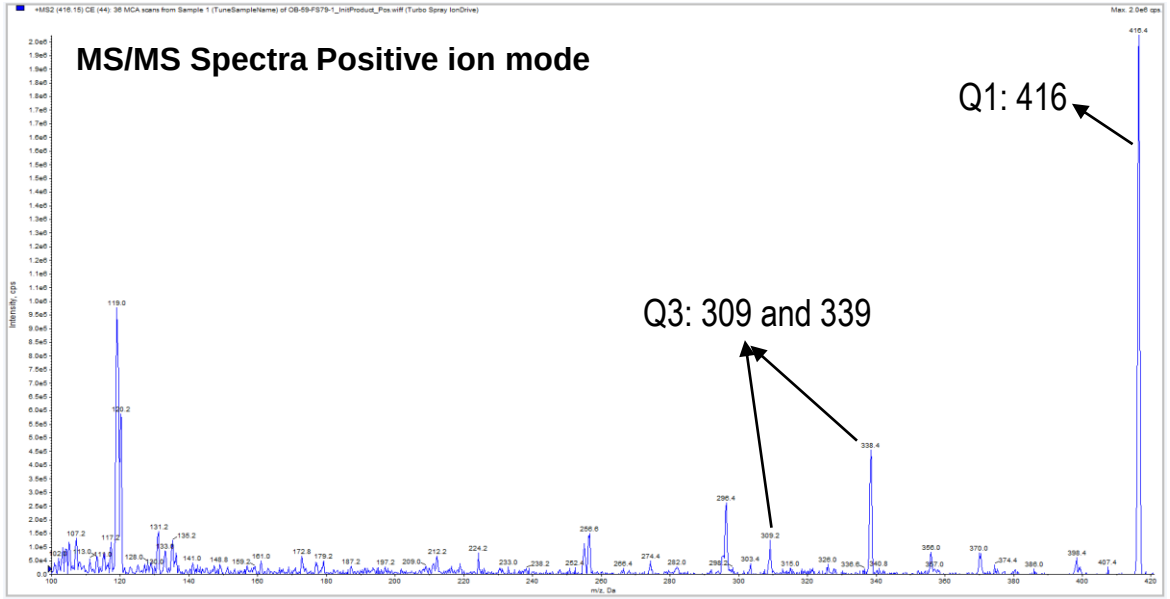
PK Bioanalysis: Case study: 01

The Problem

- Blank interference was observed for molecule, OB-59-FS79 using positive mode ions (m/z: Q1: 416 and Q3: 309 and 339)
- The ions generated in positive mode were false positive, hence interference was observed

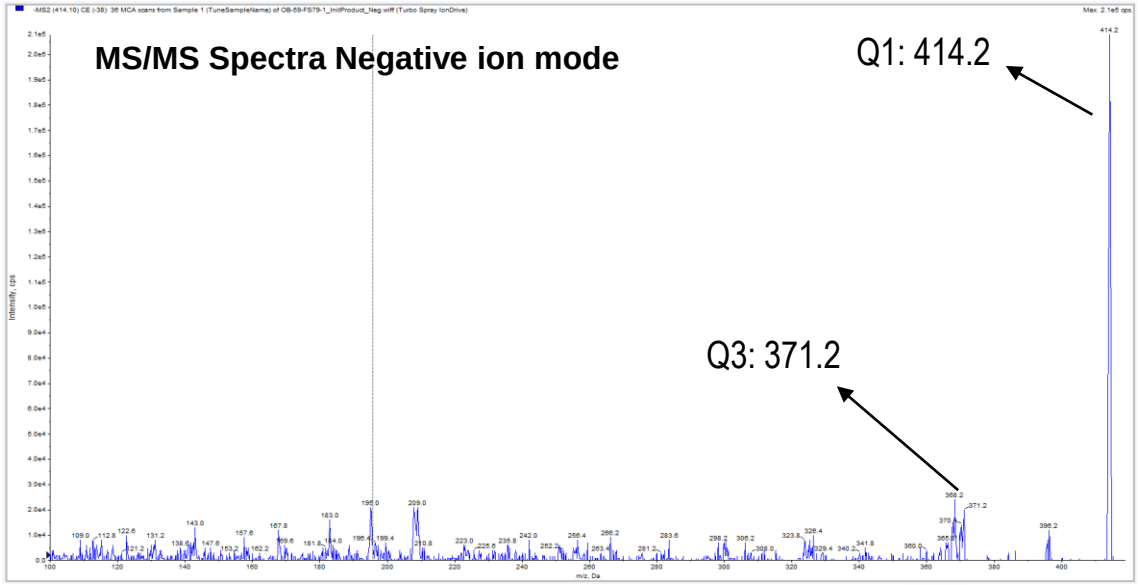
Approach

- Obtained structural information of the molecule from Med Chem & LCMS method was re-optimized as per Chemistry



Resolution

- The LCMS method molecule was modified based on the molecular structure
- The method (m/z: Q1: 414 and Q3: 371.2) was optimized in negative mode with no interference and desired sensitivity (0.1 ng/mL)



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PK Bioanalysis: Case study: 02

The Problem

- Carry-over was observed with following gradient for few molecules.
- Many injections required before study samples to eliminate carry-over

Time (mins)	Mobile phase A	Mobile Phase B
0.00	90	10
0.4	90	10
1.5	15	85
4.5	15	85
4.7	90	10
5.0	90	10

Approach

- Increasing washing cycles
- Optimization of gradient by introducing additional washing with organic solvent

Resolution

- The optimized gradient eliminated the carry over for this class of molecules with lesser number of injections

Time (mins)	Mobile Phase A	Mobile Phase B
0.00	90	10
0.4	90	10
1.5	15	85
4.0	15	85
4.01	**5**	**95**
4.50	**5**	**95**
4.51	90	10
5.0	90	10

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